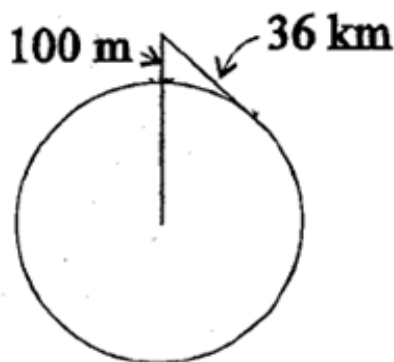
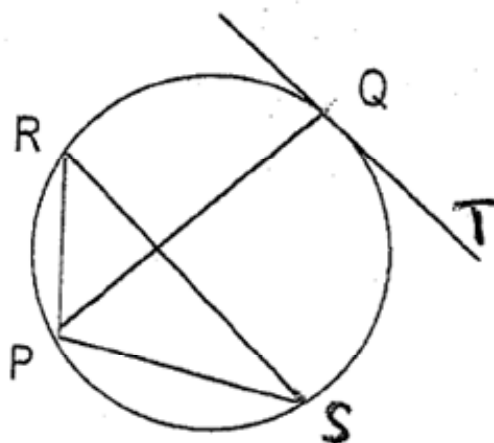


Question 1 (Start a new Booklet)**(16 marks)**

- a) Solve the inequality $\frac{x-1}{x+1} \geq 2$ [3]
- b) The point C $(-6,1)$ divides the interval AB externally in the ratio 3:1. If A has co-ordinates $(0,4)$ find the co-ordinates of B. [3]
- c) From a cliff 100 metres high, the straight line distance to the horizon is 36 kilometres. Calculate the radius of the earth to the nearest kilometre. [3]



- d) Find the obtuse angle between the lines $\frac{x}{7} + \frac{y}{5} = 1$ and $2x - 3y + 4 = 0$ to the nearest minute. [3]
- e) The diagram shows a circle with a chord PQ and another chord RS which is parallel to the tangent at Q. Prove that the chord PQ bisects $\angle RPS$. [4]



Question 2 (Start a new Booklet)**(15 marks)**

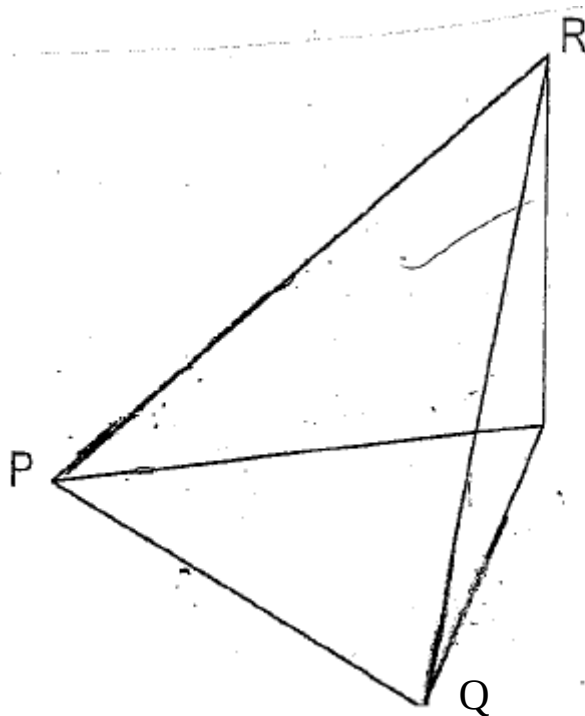
- a)** Consider the polynomial $P(x) = x^3 - 5x + c$
- i) Find the value of c if $x + 2$ is a factor of $P(x)$. [1]
 - ii) For this value of c , find $Q(x)$ such that $P(x) = (x + 2).Q(x)$. [2]
- b)** If $y = P(x)$ is an odd polynomial passing through $(8, -3)$ find the remainder when $y = P(x)$ is divided by $(x + 8)$. [2]
- c)** The quadratic equation $x^2 + 6x + c = 0$ has two real roots. These roots have opposite signs and differ by $2n$, where $n \neq 0$.
- i) Show that $n^2 = 9 - c$. [3]
 - ii) Find the set of all possible values of n . [2]
- d)** If α, β, γ are the roots of the equation $x^3 + 2x^2 - x - 5 = 0$ find
- i) $\alpha + \beta + \gamma$ [1]
 - ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ [2]
 - iii) $\alpha^2 + \beta^2 + \gamma^2$. [2]

Question 3 (Start a new Booklet)**(13 marks)**

a) Prove that $\frac{\sec^2 x}{\tan x} = \frac{1}{\sin x \cos x}$. [2]

b) Find the exact value of $\cos 15^\circ$. [2]

- c) The diagram shows a mountain of height h metres. From a point P due south the angle of elevation of summit R is found to be 14° . From another point Q , 7000 metres due east of P , the elevation of summit R is found to be 10° .



- i) Copy the diagram into your booklet and mark on all the information given. [1]
 ii) Calculate the height of the mountain to the nearest metre. [4]
- d)
- i) If $t = \tan \frac{x}{2}$ write down expressions for $\sin x$ and $\cos x$ in terms of t . [1]
 ii) Hence solve $4\cos x - 7\sin x = 1$ for $0^\circ \leq x \leq 360^\circ$ correct to the nearest minute. [3]

Question 4 (Start a new Booklet)**(12 marks)**

- a)
- i) Show that $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$. [2]
 - ii) Hence solve the equation $3\sin \theta - 4\sin^3 \theta = -1$ for $0 \leq \theta \leq 2\pi$. [2]
- b) Factorise $2\cos x \sin x - 2\sin x - \sqrt{3}\cos x + \sqrt{3}$. Hence find the solutions to $2\cos x \sin x - 2\sin x - \sqrt{3}\cos x + \sqrt{3} = 0$ in general form using radian measure. [3]
- c) For the expression $6\cos x - 8\sin x$
- i) Express this in the form $R\cos(x + \alpha)$ where α is an acute angle. [2]
- Use this relationship to find
- ii) the maximum value of $6\cos x - 8\sin x$. [1]
 - iii) the value of x to the nearest minute, for $0^\circ \leq x \leq 360^\circ$, when $6\cos x - 8\sin x = 5$. [2]

Comments

QUESTION 1.

1) $\frac{x-1}{x+1} \geq 2 \quad x \neq -1$

$(x+1)^2 \frac{x-1}{x+1} \geq 2(x+1)^2$

$x^2 - 1 \geq 2(x^2 + 2x + 1)$

$x^2 - 1 \geq 2x^2 + 4x + 2$

$0 \geq x^2 + 4x + 3$

$(x+3)(x+1) \leq 0$

$-3 \leq x < -1$ ✓

✓ (both solutions)

2) A(0, 4) B(x, y) C(-6, 1)

k: l
-3: 1

$x = \frac{kx_2 + lx_1}{k+l}$

$y = \frac{ky_2 + ly_1}{k+l}$

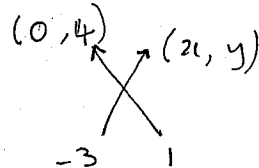
$-6 = \frac{-3x + 0}{-2}$

$1 = \frac{-3y + 4}{-2}$ ✓

$12 = -3x$
 $x = -4$ ✓

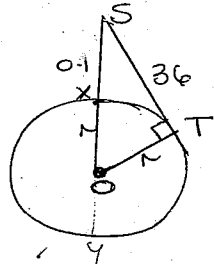
$-2 = -3y + 4$
 $3y = 6$
 $y = 2$ ✓

$\therefore B(-4, 2)$



$x = -6, y = 1$

PYTHAGORAS



$\angle OTS = 90^\circ$

(angle between radius + tangent)

$x^2 + 36^2 = (x+0.1)^2$

(using pythagoras) (with reason)

$x^2 + 36^2 = x^2 + 0.2x + 0.01$

$x = \frac{36^2 - 0.01}{0.2}$

$x = 6479.95$

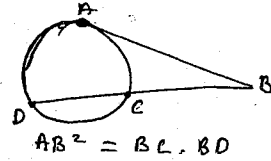
$x = 6480$

$\therefore$ the radius of the earth 6480 km.

Note Can also use property

$ST^2 = SX \cdot SY$
 $36^2 = 0.1 \cdot (0.1 + 2x) \Rightarrow x = 6480$

Property



(d) $\frac{x}{7} + \frac{y}{3} = 1 \quad 2x - 3y + 4 = 0$

$5x + 7y - 35 = 0$

$y = \frac{2x + 4}{3}$

$y = \frac{-5x + 35}{7}$

$m_2 = \frac{2}{3}$

$m_1 = -\frac{5}{7}$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{-\frac{5}{7} - \frac{2}{3}}{1 - \frac{5}{7} \times \frac{2}{3}} \right|$

✓ Correct substitution

$= \left| \frac{-\frac{29}{21}}{1 - \frac{10}{21}} \right|$

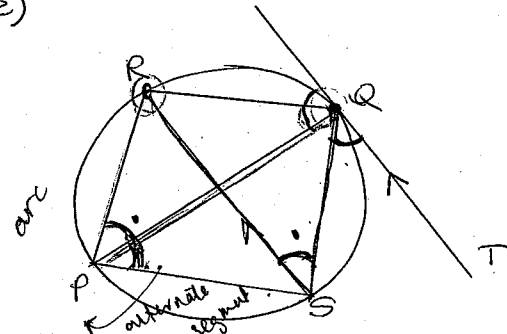
$\theta = 69^\circ 14'$

✓ acute angle

$\therefore$ Obuse angle = $110^\circ 46'$

Obuse

(e)



Join RQ & SQ.

$\angle TQS = \angle QPS$ (L between tangent + chord equals L in alternate segment)

$\angle TQS = \angle QSR$ (alt. L's, $OT \parallel RS$ give)

$\angle QSR = \angle RPS$ (L's standing on the same arc PS)

$\therefore \angle TQS = \angle RPS$ (both equal to $\angle QSR$)

$\therefore PQ$ bisects $\angle RPS$.

By construction:

RQ and SQ

Question 2,

$$1) P(x) = x^3 - 5x + c$$

$$P(-2) = -8 + 10 + c = 0 \text{ as } (x+2) \text{ is a factor.}$$

$$\therefore c = -2$$

$$ii) P(x) = x^3 - 5x - 2$$

OR

$$\begin{array}{r} x^3 - 5x - 2 \\ x+2 \overline{) x^3 - 2x - 1} \\ \underline{x^3 + 2x^2} \\ -2x^2 - 5x - 2 \\ \underline{-2x^2 - 4x} \\ -x - 2 \\ \underline{-x - 2} \\ 0 \end{array}$$

division $2b = -4$
for process $b = -2$

$$P(x) = x^3 - 5x - 2$$

$$P(x) = (x+2)(x^2 + bx - 1)$$

$$2b - 1 = -5$$

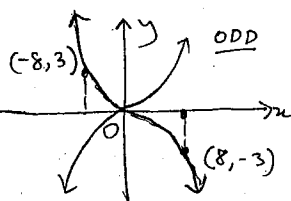
$$\therefore P(x) = (x+2)(x^2 - 2x - 1)$$

OR use Remainder theorem:

ODD: reflected on y-axis
Hence: origin

$$y = 3$$

relationship



$$\therefore P(x) = (x+2)(x^2 - 2x - 1)$$

$$\therefore Q(x) = x^2 - 2x - 1$$

$$(b) y = P(x)$$

$$\text{thru } (8, -3) \Rightarrow P(8) = -3$$

$$\text{for an odd fn } P(x) = -P(-x)$$

$$\therefore -P(-8) = -3$$

$$\therefore P(-8) = 3$$

The remainder when $P(x)$ is divided by $(x+8)$ is given by $P(-8)$.

$$\therefore \text{remainder} = 3$$

$$(c) x^2 + 6x + c = 0$$

$$\therefore a=1, b=6, c=c$$

let the roots be $\alpha, \alpha+2n$

$$\text{sum, } \alpha + \alpha + 2n = -\frac{b}{a}$$

$$2\alpha + 2n = -6$$

$$\alpha + n = -3$$

$$\alpha = -3 - n$$

$$Q2 (c) (cont'd) x^2 + 6x + c = 0$$

$$\text{Also } \alpha(\alpha+2n) = \frac{c}{a}$$

$$(-3-n)(-3-n+2n) = c$$

$$(-3-n)(-3+n) = c$$

$$9 - n^2 = c$$

$$\therefore n^2 = 9 - c \text{ as req'd.}$$

$$c = 9 - n^2$$

ii) Since roots are opposite in sign $\Rightarrow$ product is -ve.

$$\therefore c < 0$$

reasoning.

$$\therefore 9 - n^2 < 0 \text{ (from above)}$$

$$(3-n)(3+n) < 0$$

$$\{n: n < -3, n > 3\}$$



$$(d) x^3 + 2x^2 - x - 5 = 0$$

$$a=1 \quad b=2 \quad c=-1 \quad d=-5$$

$$i) \alpha + \beta + \gamma = -\frac{b}{a} = -2$$

$$ii) \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{-1}{5}$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = -1$$

$$\alpha\beta\gamma = -\frac{d}{a} = 5$$

$$iii) \alpha^2 + \beta^2 + \gamma^2$$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= 4 - 2(-1)$$

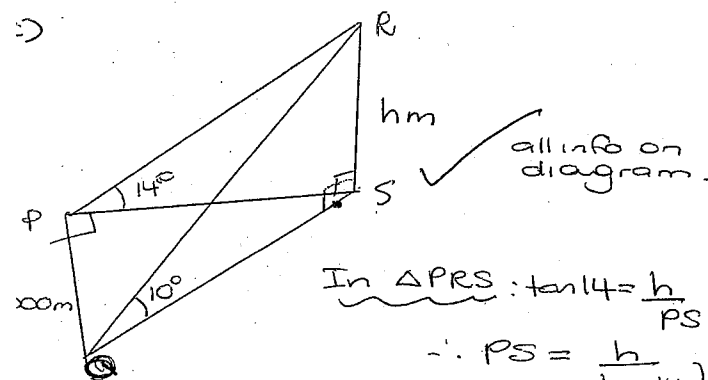
$$= 6$$

QUESTION 3

$$2) \frac{\sec^2 x}{\tan x} = \frac{1}{\sin x \cos x}$$

$$\begin{aligned} \text{L.H.S} &= \frac{1}{\cos^2 x} \times \frac{1}{\tan x} \\ &= \frac{1}{\cos^2 x} \times \frac{\cos x}{\sin x} \quad \checkmark \text{ substitution} \\ &= \frac{1}{\cos x \sin x} \quad \checkmark \text{ simplifying.} \\ &= \text{R.H.S.} \end{aligned}$$

$$\begin{aligned} 3) \cos 15^\circ &= \cos(45^\circ - 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \quad \checkmark \text{ expansion.} \\ &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}} \left(\times \frac{\sqrt{2}}{\sqrt{2}} \right) \quad \checkmark \text{ either} \\ &= \frac{\sqrt{6}+\sqrt{2}}{4} \end{aligned}$$



$$\begin{aligned} \text{In } \triangle PRS : \tan 14^\circ &= \frac{h}{PS} \\ \therefore PS &= \frac{h}{\tan 14^\circ} \\ \text{In } \triangle RQS : \tan 10^\circ &= \frac{h}{QS} \\ QS &= \frac{h}{\tan 10^\circ} \end{aligned}$$

In $\triangle PQS$

$$QS^2 = 7000^2 + PS^2 \quad (\text{using pythagoras})$$

$$\frac{h^2}{\tan^2 10^\circ} = 7000^2 + \frac{h^2}{\tan^2 14^\circ} \quad \checkmark$$

$$\frac{h^2}{\tan^2 10^\circ} - \frac{h^2}{\tan^2 14^\circ} = 7000^2$$

$$h^2 \left(\frac{1}{\tan^2 10^\circ} - \frac{1}{\tan^2 14^\circ} \right) = 7000^2$$

$$h^2 = \frac{7000^2}{\frac{1}{\tan^2 10^\circ} - \frac{1}{\tan^2 14^\circ}} \quad \checkmark \text{ for expression in } h^2$$

$$h = 1745.8 \dots$$

The height of mountain is 1746 m (nearest m). $\checkmark$ answer

$$(d) t = \tan \frac{x}{2}$$

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

both $\checkmark$

$$4 \cos x - 7 \sin x = 1 \quad 0^\circ \leq x \leq 360^\circ$$

$$4 \left(\frac{1-t^2}{1+t^2} \right) - 7 \left(\frac{2t}{1+t^2} \right) = 1$$

$$4 - 4t^2 - 14t = 1 + t^2$$

$$5t^2 + 14t - 3 = 0$$

$$t = \frac{-14 \pm \sqrt{14^2 - 4 \times 5 \times (-3)}}{2 \times 5}$$

$$= \frac{-14 \pm 16}{10}$$

$$t = -3, \frac{1}{5}$$

$$\tan \frac{x}{2} = -3, \frac{1}{5}$$

$$\text{acute } \frac{x}{2} = 71^\circ 34'$$

$$x = 180 - 71^\circ 34' = 108^\circ 26'$$

$$x = 216^\circ 52'$$

both solns for t. $\checkmark$

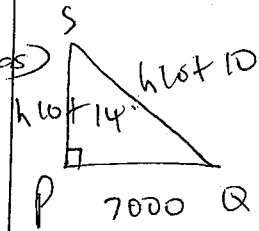
$$0^\circ \leq x < 360^\circ$$

$$0^\circ \leq \frac{x}{2} \leq 180^\circ$$

$$\tan \frac{x}{2} = \frac{1}{5}$$

$$\frac{x}{2} = 11^\circ 19'$$

$$x = 22^\circ 37'$$



$$\begin{matrix} 5t & -1 \\ t & 3 \end{matrix}$$

$$5t^2 + 14t - 3 = 0$$

$$(5t-1)(t+3) = 0$$

QUESTION 4

a) (i) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

L.H.S = $\sin(2\theta + \theta)$

$= \sin 2\theta \cos \theta + \sin \theta \cos 2\theta$

$= 2\sin\theta \cos^2\theta + \sin\theta(\cos^2\theta - \sin^2\theta)$

$= 2\sin\theta \cos^2\theta + \sin\theta(1 - \sin^2\theta) - \sin^3\theta$

$= 2\sin\theta(1 - \sin^2\theta) + \sin\theta - \sin^3\theta - \sin^3\theta$

$= 2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta$

$= 3\sin\theta - 4\sin^3\theta$

$= R.H.S.$

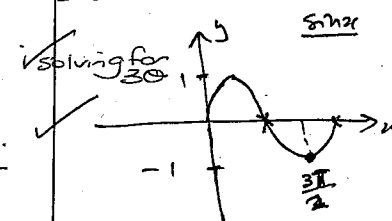
i) $3\sin\theta - 4\sin^3\theta = -1$

$\sin 3\theta = -1$

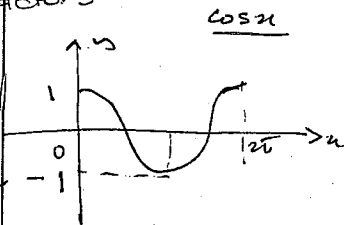
$3\theta = \frac{3\pi}{2}, \frac{7\pi}{2}, \frac{11\pi}{2}$

$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

$0 \leq 3\theta \leq 6\pi$



correct factors



ii) $2\cos x \sin x - 2\sin x - \sqrt{3}\cos x + \sqrt{3}$

$= 2\sin x(\cos x - 1) - \sqrt{3}(\cos x - 1)$

$= (2\sin x - \sqrt{3})(\cos x - 1)$

$(2\sin x - \sqrt{3})(\cos x - 1) = 0$

$2\sin x - \sqrt{3} = 0$

$\sin x = \frac{\sqrt{3}}{2}$

$\sin x = \sin \frac{\pi}{3}$

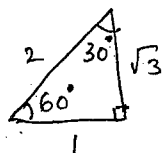
$x = n\pi + (-1)^n \frac{\pi}{3}$

$\cos x = 1$

$\cos x = 0, 2\pi$

$x = 2n\pi$

$x = 2\pi n$



Question 4 (cont'd.)

(c) (i) $6\cos x - 8\sin x \equiv R\cos(x+\alpha)$

$R = \sqrt{6^2 + 8^2}$

$= 10$

$\therefore 6\cos x - 8\sin x \equiv 10\cos(x+\alpha)$

$\frac{3}{5}\cos x - \frac{4}{5}\sin x \equiv \cos(x+\alpha)$

$\equiv \cos x \cos \alpha - \sin x \sin \alpha$

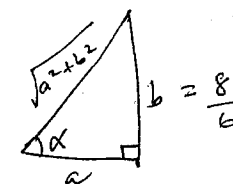
$\therefore \cos \alpha = \frac{3}{5}$

$\sin \alpha = \frac{4}{5}$

$\Rightarrow \tan \alpha = \frac{4}{3}$

$\alpha = 53^\circ 8'$

resolving α



$\tan \alpha = 53^\circ 8'$

$\therefore 6\cos x - 8\sin x \equiv 10\cos(x+53^\circ 8')$

ii) Max value = 10

iii) $6\cos x - 8\sin x = 5$

$\therefore 10\cos(x+53^\circ 8') = 5$

$\cos(x+53^\circ 8') = \frac{1}{2}$

acute $(x+53^\circ 8') = 60^\circ$

$x + 53^\circ 8' = 60^\circ, 300^\circ$

$x = 6^\circ 52', 246^\circ 52'$